

Indraprastha Institute of Information Technology Delhi (IIITD)
Department of Computational Biotechnology
BIO213 – Introduction to Quantitative Biology
MID-SEM EXAM (September 28, 2025)

Time duration: 1 hour

Total marks: 50

SECTION I

Multiple choice questions.

(1 mark each)

Question 1. In BLAST, the E-value indicates:

- i. The probability of the query being an exact match
- ii. The expected number of random hits of a given score
- iii. The evolutionary distance between two sequences
- iv. The percentage of identical residues

Question 2. Which of the following errors occurs when two fragments from different regions of DNA join end-to-end?

- i. Contamination
- ii. Base call error
- iii. Chimera
- iv. Deletion

Question 3. Which statistical method evaluates the reliability of a phylogenetic tree by resampling with replacement?

- i. Jackknifing
- ii. Bootstrapping
- iii. UPGMA
- iv. Maximum Parsimony

Question 4. Which one is true for a Eulerian Walk in De Bruijn graph assembly?

- i. It visits every vertex exactly once
- ii. It visits every edge exactly once
- iii. It minimizes sequence length
- iv. It removes repeats automatically

Question 5. You are aligning two proteins where only the central domain is conserved, but both sequences have long, unrelated N- and C-terminal tails. Which alignment method prevents those tails from dragging down the score?

- i. Global
- ii. Local
- iii. End-free
- iv. Multiple sequence alignment

Question 6. You observe a dot-plot diagonal that shifts slightly downwards midway, as if “broken.” What does this pattern suggest?

- i. An inversion
- ii. Complete unrelatedness
- iii. A small insertion/deletion event
- iv. Two independent alignments

Question 7. In the MULTICONTIG model, what is a "weakest link"?

- i. The fragment with the most errors.
- ii. The smallest link in a layout, defined by the number of overlapping positions.
- iii. The longest fragment in a set.
- iv. A fragment that does not overlap with any other fragment.

Question 8. Consider the following multiple sequence alignment layout

- | | |
|-------|--------|
| i. KK | ii. KH |
| KH | KR |
| RK | KH |
| HR | KR |
| NK | KR |
| KA | HK |

Which of the above (i or ii) is more likely to have a higher log-odd ratio for the event $K \rightarrow R$?

Question 9. Off-springs in Genetic algorithm are generated using

- i. Mutations and crossover
- ii. Natural selection and crossover
- iii. Natural selection and mutations

Question 10. Which of the following is an example of transition and transversion respectively?

- i. $A \rightarrow C$ and $A \rightarrow T$
- ii. $C \rightarrow T$ and $G \rightarrow A$
- iii. $G \rightarrow C$ and $C \rightarrow T$
- iv. $A \rightarrow G$ and $G \rightarrow T$

SECTION II

Answer the following questions (do any 4)

(5 marks each)

Question 11. In the RECONSTRUCTION model for fragment assembly, if $\epsilon = 0.3$, is the placement of fragment GTCCA valid if its edit distance to the consensus sequence is 2?

Question 12. (a) Why is Neighbor Joining (NJ) often preferred over Unweighted Pair Group Method with Arithmetic Mean (UPGMA) for phylogenetic tree over UPGMA?

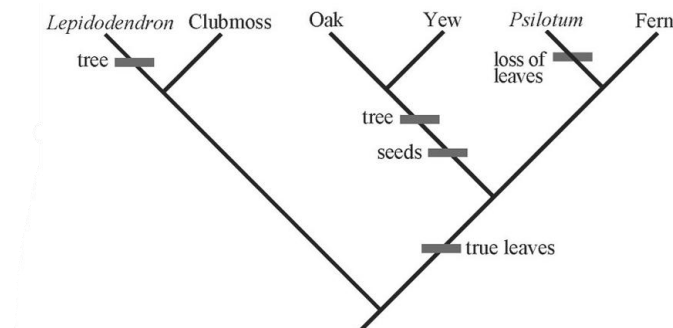
(b) In NJ, if $d(A,B) = 4$ and $r(A)+r(B) = 12$, $N=4$, what is $M(AB)$?

Question 13. A researcher is sequencing a bacterial genome of size 3 Mbp. They want to achieve a minimum of 12x coverage using reads of length 150 bp.

(a) How many bases of sequence data are required in total?

(b) How many reads of length 150 bp must be generated to reach the target coverage?

Question 14. In the tree shown below, assume that the ancestor was an herb (not a tree) without leaves or seeds. Based on the tree and assuming that all evolutionary changes in these traits are shown, which of the tips has a tree habit and lacks true leaves?



- i. *Lepidodendron*
- ii. Clubmoss
- iii. Oak
- iv. *Psilotum*
- v. Fern

Question 15. Construct an overlap graph for $F = (CTGA, GACT, TGAC, ACTG)$ with 3-mer overlaps, and derive the shortest common superstring.

SECTION III

Long answer questions (do any 2)

(10 marks each)

Question 15. Find the local regions of similarity between the following DNA sequences using dynamic programming and the given scoring scheme.

DNA sequences: (1) TGTTACGG and (2) GGTTGACTA

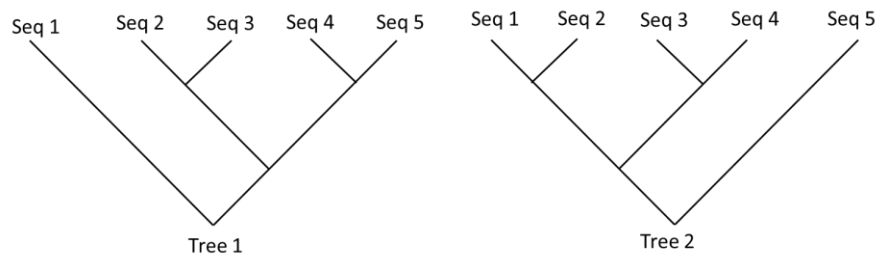
Scoring function: Match = +3, Mismatch = -3, Gap = -2.

Question 16. A researcher is working with the following sequences of flagellin proteins from six different bacteria of similar lineage to find the phylogenetic relationship between.

	1	2	3	4	5	6	7	8	9
Seq 1	A	T	T	T	A	A	T	A	C
Seq 2	A	T	G	T	C	A	T	A	A
Seq 3	G	A	A	G	A	A	T	A	T
Seq 4	G	C	G	A	G	A	C	C	C
Seq 5	G	C	G	T	C	A	A	T	G

(i) Identify the parsimony informative sites from the given alignment layout?

(ii) Using the informative sites, determine the minimum number of mutations required for each of the following unrooted trees?



(iii) Which tree is favorable by maximum parsimony principle?

Question 17. Given a set of reads comprising of ATGC, GCGT, TGCG and CGTA,

- List all k-mers of length 3 from these reads.
- Construct the De Bruijn graph with (k-1)-mers as nodes.
- Find a Eulerian path that reconstructs the genome.